Algebra II with Trigonometry Chapter 5.5

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4465374_alg-2trig-h-chp-5-5-note-docx-1.pdf

Problems

1. In triangle ABC, $A = 30^\circ$, $a = 10$, and $b = 10\sqrt{3}$. Determine how many triangles are possible, and for each one find B and C.

2. Triangle ABC satisfies $A = 15^\circ$, $B = 45^\circ$, and $a + b = 4\sqrt{2}$. Find c.

3. In triangle ABC, $a = 7$, $b = 14$, and $A = 30^\circ$. Without solving the whole triangle, find the two possible values of $B - C$.

4. Two observers A and B stand 60 m apart on level ground. A balloon is on the same side of both observers. The angles of elevation are 75° from A and 45° from B. Find the height of the balloon.

5. In triangle ABC, $A = 20^\circ$ and $a = b - c$. Find $B - C$.

6. A point D lies on side AB of triangle ABC with $CD = CA$. Also $BC = CD$, $CA = 28$, and $\angle A = 30^\circ$. Find $\angle DCB$.